



Exploratory Data Analysis (EDA)

Mini Project Report



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Business Challenge

In a highly competitive online grocery market, companies like BigBasket operate at scale while working with thin margins. Managing thousands of SKUs across multiple categories makes pricing and discount decisions complex and high-impact.

The core question is not just sales — it is profitability.

- Are discounts increasing sustainable revenue, or simply driving volume at the expense of margins?
- Do higher-rated products consistently outperform lower-rated ones?
- Which categories generate the most value, and which underperform despite heavy promotions?

To answer these questions, this analysis examines pricing patterns, discount impact, rating distribution, and category-wise performance trends within BigBasket's product dataset.

The objective is to translate raw data into actionable insights that support strategic pricing, smarter inventory management, and stronger competitive positioning.



What the Data Contains?

The dataset includes 10 key attributes describing each product on the platform, covering identification, pricing, classification, and customer feedback. These variables form the foundation for structured business analysis.

Product Identification

- index
- product
- brand
- description

Pricing Attributes

- sale_price
- market_price

Classification & Feedback

- category
- sub_category
- type
- rating

The availability of both market price and sale price enables discount analysis, while ratings provide insight into customer satisfaction. Category and brand information help identify distribution patterns and performance trends across the platform.

Making the Data Ready: Initial Inspection

The dataset contains 27,555 records and 10 attributes, including both numerical and categorical features. An initial inspection was performed to review data types, non-null counts, and memory usage, helping identify missing values and structural patterns before further analysis.

Key Observations

- The dataset contains 27,555 records and 10 attributes.
- Both numerical and categorical features are present.
- Most columns have complete data.
- The rating column contains 8,636 missing values.
- Minor missing values exist in brand and description fields.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   index           27555 non-null  int64  
1   product         27554 non-null  object  
2   category        27555 non-null  object  
3   sub_category    27555 non-null  object  
4   brand           27554 non-null  object  
5   sale_price      27549 non-null  float64 
6   market_price    27555 non-null  float64 
7   type            27555 non-null  object  
8   rating          18919 non-null  float64 
9   description     27440 non-null  object  
dtypes: float64(3), int64(1), object(6)
memory usage: 2.1+ MB
```

DataFrame Structure (df.info())

```
df.isnull().sum()
```

```
index           0
product         1
category        0
sub_category    0
brand           1
sale_price      6
market_price    0
type            0
rating          8636
description     115
dtype: int64
```

Missing Value Summary (df.isnull().sum())

These observations guided the data cleaning and transformation steps in the next phase of analysis.

Making the Data Ready: Data Cleaning and Treatment

1. Removing Irrelevant Columns

To prepare the dataset for analysis, several structured cleaning steps were performed.

First, unnecessary columns were removed to reduce noise and keep the dataset focused on meaningful variables.

The index column was dropped because it only acted as a row identifier and did not contribute to business analysis.

The description column was also removed, as it contained textual information that was not required for numerical and category-based analysis.

```
df.drop(columns=["description","index"],inplace= True)
```

2. Completing Missing Details

Next, missing values in categorical fields such as product and brand were handled.

Instead of replacing them with the most frequent value (mode), which could introduce incorrect assumptions, these missing entries were filled with “Unknown Product” and “Unknown Brand” to maintain data transparency without distorting brand or product distribution.

```
df["product"].fillna("Unknown Product",inplace = True)
```

```
df["brand"].fillna("Unknown Brand",inplace = True)
```

3. Stabilizing Rating Values

For the rating column, missing values were filled using the mean.

Since ratings are numerical and were observed to have a reasonably balanced distribution within a fixed range, mean imputation helped preserve the overall central tendency without significantly altering the data pattern.

```
df["rating"].fillna(df["rating"].mean(),inplace= True)
```

4. Refining Sale Price Values

In the case of sale_price, missing values were treated using the median price within each category.

Price data is often skewed due to the presence of premium or bulk products, and the median is less affected by extreme values.

Using category-wise median ensured that imputed prices remained realistic and consistent within their respective product groups.

```
df['sale_price'] = df['sale_price'].fillna(  
    df.groupby('category')['sale_price'].transform('median'))
```

- These preprocessing steps ensured a structured, consistent, and analysis-ready dataset for subsequent visualization and insight generation.

Pricing Structure Analysis

1. What the Pricing Metrics Show

df.describe()				
	index	sale_price	market_price	rating
count	27555.00000	27555.000000	27555.000000	27555.000000
mean	13778.00000	334.615556	382.056664	3.943295
std	7954.58767	1201.973950	581.730717	0.612516
min	1.00000	2.450000	3.000000	1.000000
25%	6889.50000	95.000000	100.000000	3.943295
50%	13778.00000	190.080000	220.000000	3.943295
75%	20666.50000	359.000000	425.000000	4.200000
max	27555.00000	112475.000000	12500.000000	5.000000

- **Competitive Discounting:** Products are generally priced below market averages, with a mean sale price of ₹334.61 vs. a ₹382.05 market benchmark.
- **Segment Concentration:** The core catalog sits in the ₹95–₹359 bracket, though high price variance confirms a broad reach from economy to luxury tiers.
- **Quality Consistency:** A narrow variance in the 3.94 average rating demonstrates that customer satisfaction is decoupled from price, indicating high perceived value across the board.

2. Inconsistency Between Sale Price and Market Price

```
df[df["sale_price"] > df["market_price"]].shape
(10, 8)
```

The audit identified 10 instances where the Sale Price exceeds the Market Price, violating standard retail logic ($\text{Sale Price} \leq \text{Market Price}$).

Critical Observations:

- **Data Irregularity:** Presence of "negative discounts" suggests manual entry errors or outdated market price benchmarks.
- **Analytical Risk:** These outliers distort discount averages and compromise the accuracy of margin reporting.
- **Recommendation:** Immediate validation and correction of flagged SKUs are required to maintain dataset reliability.

3. Removing Records with Pricing Inconsistency

```
df = df[df["sale_price"] <= df["market_price"]]
```

After identifying the pricing inconsistency, the dataset was cleaned by removing records where the sale price was greater than the market price. Since a discounted price should not exceed the original market price, such entries were considered logically incorrect.

A filtering condition was applied to retain only those records where the sale price is less than or equal to the market price. After applying this step, the dataset was checked again and no inconsistent records remained. This confirms that the pricing structure is now logically valid and suitable for further analysis.

4. Confirmation of Issue Resolution

```
df[df["sale_price"] > df["market_price"]].shape  
  
(0, 8)
```

To confirm that the pricing issue was successfully resolved, the dataset was examined again after applying the cleaning condition. Initially, the dataset showed 10 inconsistent records (10, 8) where the sale price exceeded the market price.

After removing these invalid entries, the condition was rechecked and the output showed (0, 8). This confirms that no mismatched pricing records remain in the dataset, indicating that the issue has been fully resolved and the data is now logically consistent.

Which Products Were Sold the Most?

```
df['product'].value_counts().head(10)
```

```
product
Turmeric Powder/Arisina Pudi      26
Extra Virgin Olive Oil            15
Cow Ghee/Tuppa                   14
Soft Drink                       12
Olive Oil - Extra Virgin          12
Colorsilk Hair Colour With Keratin 12
Coriander Powder                  11
Powder - Coriander                11
Ghee/Tuppa                       11
Casting Creme Gloss Hair Color    10
Name: count, dtype: int64
```

To identify the most sold products, the frequency of each product in the dataset was calculated using value counts.

This shows how many times each product appears, which represents how often it was sold.

The analysis reveals that products such as Turmeric Powder/Arisina Pudi, Extra Virgin Olive Oil, and Cow Ghee/Tuppa dominate sales frequency, indicating strong customer demand and consistent purchase behavior.

Which Products Were Sold the Least?

```
df['product'].value_counts().tail(10)
```

```
product
Shot Body Spray - Red Stallion      1
Face Wash - Oil Control, Active     1
Mold & Mildew Remover with Bleach   1
Just Spray - Mosquito Repellent Room Spray 1
Dove Plastic Soap Case - Assorted Colour 1
Detoxifying Mineral Pack           1
Salted Pumpkin                     1
Plastic Hook/Hanger Card Self Adhesive/Stickable - BH 048 1
Organic Tofu - Soy Paneer           1
Mutton - Boneless, 2-3 pcs, Antibiotic Residue-Free, Growth Hormone-Free 1
Name: count, dtype: int64
```

To determine the least sold products, the dataset was sorted in ascending order based on product frequency.

The output shows several products with a count of 1, meaning they appear only once in the dataset. This indicates that these products were sold very few times compared to others. Such low frequency may suggest lower demand or limited transactions within the dataset.

How the Discount Amount Was Calculated?

```
df["discount_amount"] = df["market_price"] - df["sale_price"]  
print(df["discount_amount"])
```

```
0          0.00  
1          0.00  
2        131.00  
3         27.00  
4          0.00  
...  
27550       49.80  
27551        7.50  
27552        0.00  
27553       99.00  
27554      175.47  
Name: discount_amount, Length: 27545, dtype: float64
```

A new column, `discount_amount`, was created to calculate the monetary discount for each product.

The discount was computed by subtracting the sale price from the market price for each product (Market Price – Sale Price).

This calculation gives the actual monetary discount offered on each item.

The output confirms the calculated discount values across all 27,545 records. A value of 0 indicates no discount, while positive values represent the actual price reduction from the market price

How the Discount Percentage Was Calculated?

```
df["discount_percentage"] = ((df["market_price"] - df["sale_price"]) / df["market_price"]) * 100
print(df["discount_percentage"])

0         0.000000
1         0.000000
2        52.400000
3        15.340909
4         0.000000
...
27550     20.000000
27551     10.000000
27552      0.000000
27553     20.000000
27554     44.992308
Name: discount_percentage, Length: 27545, dtype: float64
```

The discount percentage was calculated to measure the relative price reduction compared to the original market price.

A new column named `discount_percentage` was created using the formula:

$$(\text{Market Price} - \text{Sale Price}) / \text{Market Price} \times 100$$

This calculation shows the percentage reduction applied to each product. The output displays the discount percentage for all 27,545 records in the dataset.

A value of 0% indicates no discount, while higher values represent greater price reductions. This metric helps compare discounts across products more effectively, regardless of their price differences.

Top Discounted Products

```
df.sort_values("discount_percentage", ascending=False).head(5)
```

	product	category	sub_category	brand	sale_price	market_price	type	rating	discount_amount	discount_percentage
26976	Curry Leaves	Fruits & Vegetables	Herbs & Seasonings	Fresho	2.45	15.0	Indian & Exotic Herbs	3.943295	12.55	83.666667
17713	Fruit & Vegetables Hand Juicer	Kitchen, Garden & Pets	Kitchen Accessories	Triones	349.00	1995.0	Kitchen Tools & Other Accessories	1.900000	1646.00	82.506266
13318	Small Silicone Spatula With Plastic Handle - A...	Kitchen, Garden & Pets	Bakeware	HAZEL	75.00	399.0	Baking Tools & Brushes	3.900000	324.00	81.203008
13740	Decorative Party Light Big Star String LED Lig...	Kitchen, Garden & Pets	Appliances & Electricals	Lexton	209.00	1099.0	CFL & Led Bulbs	3.200000	890.00	80.982712
10438	NHS 860 Temperature Control Professional Hair ...	Beauty & Hygiene	Hair Care	Nova	467.03	2395.0	Tools & Accessories	3.200000	1927.97	80.499791

The dataset was sorted in descending order of discount percentage to identify the most heavily discounted products.

The top five products with the highest discount percentages were then extracted for analysis.

The analysis reveals that products such as Curry Leaves (83.66%), Fruit & Vegetables Hand Juicer (82.50%), and Small Silicone Spatula (81.20%) exhibit the highest discount percentages in the dataset.

These products have significantly reduced sale prices compared to their original market prices.

This indicates that certain items are offered with very high promotional discounts, possibly to attract customers or clear inventory.

Least Discounted Products

```
df.sort_values("discount_percentage").head(5)
```

	product	category	sub_category	brand	sale_price	market_price	type	rating	discount_amount	discount_percentage
22	Chia Seeds	Foodgrains, Oil & Masala	Dry Fruits	NaturoBell	120.0	120.0	Other Dry Fruits	3.9	0.0	0.0
1	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.0	180.0	Water & Fridge Bottles	2.3	0.0	0.0
27552	Peri-Peri Sweet Potato Chips	Gourmet & World Food	Snacks, Dry Fruits, Nuts	FabBox	200.0	200.0	Nachos & Chips	3.8	0.0	0.0
4	Creme Soft Soap - For Hands & Body	Beauty & Hygiene	Bath & Hand Wash	Nivea	162.0	162.0	Bathing Bars & Soaps	4.4	0.0	0.0
27549	Papad - Garlic Disco	Snacks & Branded Foods	Ready To Cook & Eat	Atish	61.0	61.0	Papads, Ready To Fry	4.0	0.0	0.0

To identify the least discounted products, the dataset was sorted in ascending order based on discount percentage.

The bottom products were extracted, and the results show that these items have a discount percentage of 0%. This means their sale price is equal to their market price, indicating that no discount has been applied.

These products are being sold at their original listed price without any promotional reduction.

Outlier Detection Using IQR

```
Q1 = df["sale_price"].quantile(0.25)
Q3 = df["sale_price"].quantile(0.75)
print("Q1:", Q1)
print("Q3:", Q3)
```

```
Q1: 95.0
Q3: 359.0
```

```
IQR = Q3 - Q1
print(IQR)
```

```
264.0
```

To detect outliers in the sale_price column, the Interquartile Range (IQR) method was applied. First, the 25th percentile (Q1) and the 75th percentile (Q3) were calculated.

The value of Q1 is 95, which means that 25% of the products have a sale price less than or equal to ₹95.

The value of Q3 is 359, meaning that 75% of the products have a sale price less than or equal to ₹359. This also indicates that the middle 50% of the data lies between ₹95 and ₹359.

The Interquartile Range (IQR) was then calculated as $Q3 - Q1$, which equals 264. The IQR represents the spread of the central 50% of the data and is used to identify extreme values outside the normal price range.

Calculation of Outlier Boundaries

```
lower_bound = Q1 - 1.5 * IQR  
upper_bound = Q3 + 1.5 * IQR
```

```
lower_bound
```

```
np.float64(-301.0)
```

```
upper_bound
```

```
np.float64(755.0)
```

After calculating the IQR, the lower and upper boundaries were determined to identify outliers in the sale_price column.

The lower boundary was calculated using the formula: $Q1 - 1.5 \times IQR$, and the upper boundary was calculated as $Q3 + 1.5 \times IQR$. Based on the calculations, the lower boundary is -301 and the upper boundary is 755.

Any sale price below -301 or above 755 would be considered an outlier.

Since sale prices cannot realistically be negative, the main focus is on values greater than 755, which represent unusually high prices compared to the majority of the dataset.

Count of Sale Prices Above the Upper Boundary

```
df[(df["sale_price"] < lower_bound) |  
    (df["sale_price"] > upper_bound)].shape  
  
(2205, 8)
```

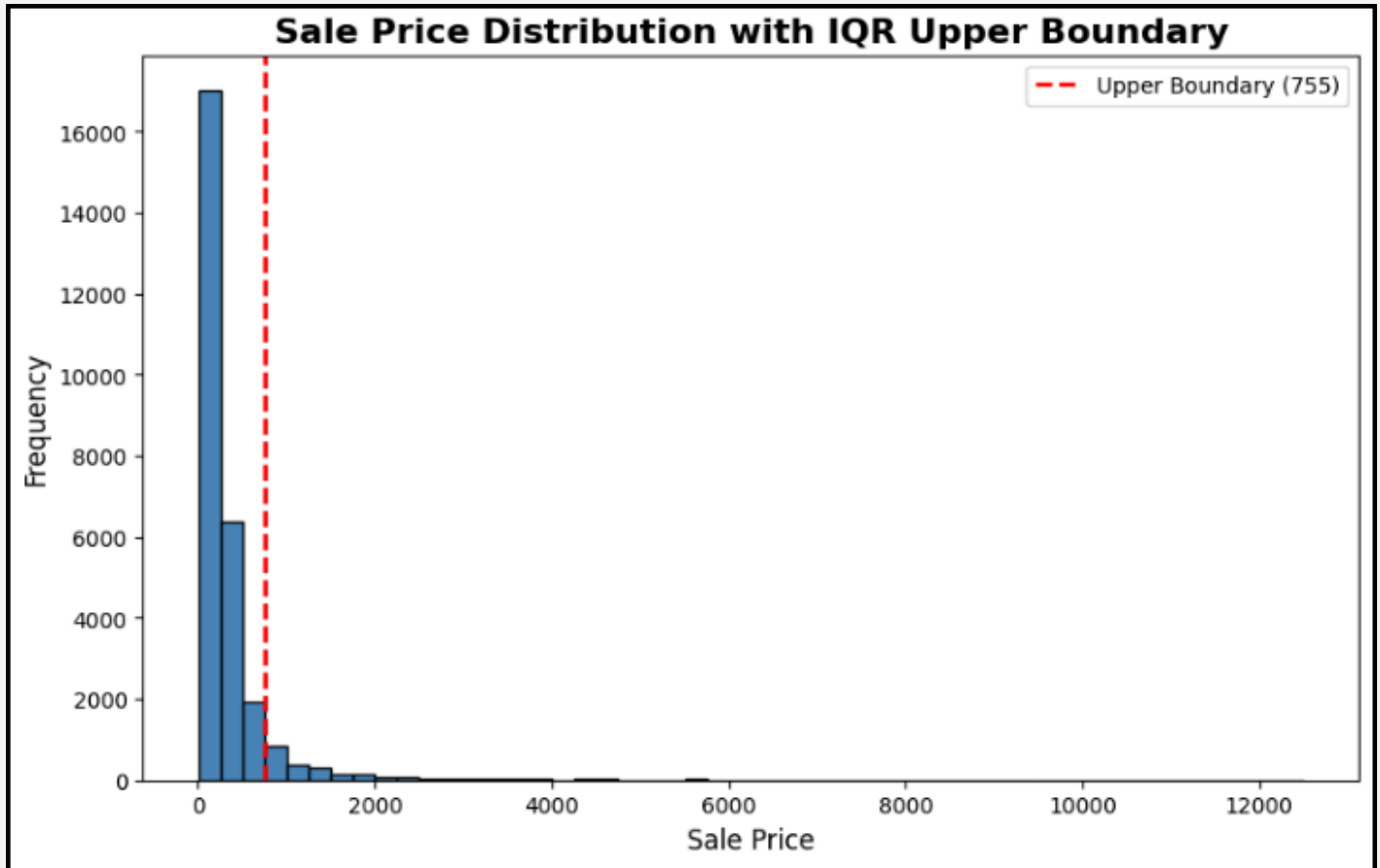
After calculating the lower and upper boundaries using the IQR method, the dataset was filtered to identify values that fall outside these limits.

The result shows (2205, 8), which means that 2,205 records in the dataset exceed the acceptable range defined by the statistical boundaries.

Since the lower boundary is negative and sale prices cannot be negative, the identified outliers are mainly due to values exceeding the upper boundary of 755.

These values are statistically considered extreme when compared to the central distribution of the data.

Visual Validation of Identified Outliers



The histogram illustrates the overall distribution of sale prices across the dataset.

The red dashed line represents the calculated IQR upper boundary at 755. While 2,205 values were statistically identified as exceeding this boundary, the plot clearly shows a right-skewed distribution with a natural long tail extending toward higher price ranges.

These higher values are not isolated or random errors; instead, they represent premium and high-priced products within the dataset.

Since the distribution pattern appears consistent and logically structured, these values are considered valid observations rather than data anomalies. Therefore, no replacement with the mean was performed.

Sale Price Distribution Pattern

```
df["sale_price"].describe()
```

```
count    27545.000000
mean      322.492820
std       486.156854
min        2.450000
25%       95.000000
50%      190.010000
75%      359.000000
max     12500.000000
Name: sale_price, dtype: float64
```

The statistical breakdown of the sale_price column provides deeper insight into the distribution pattern.

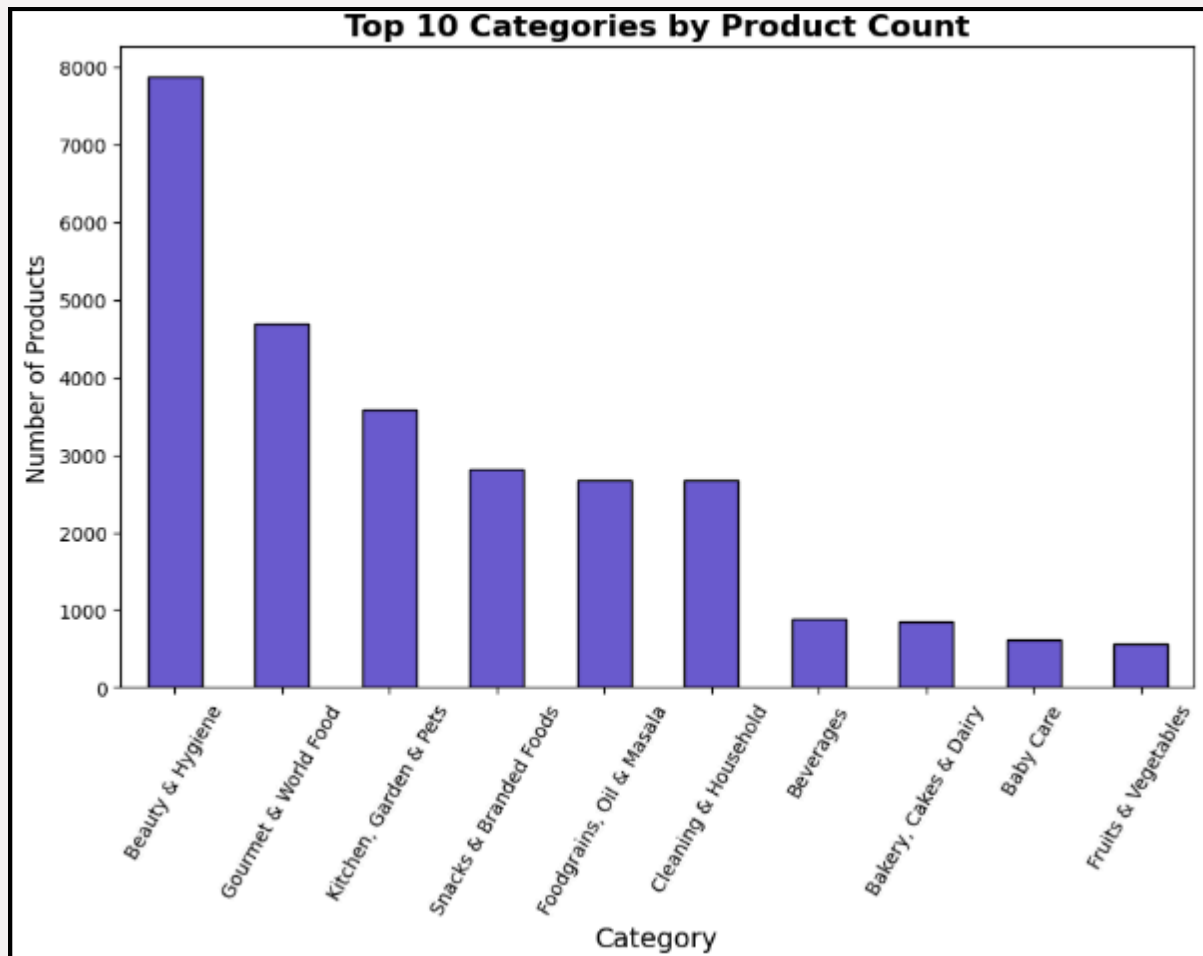
The mean sale price is ₹322, while the median (50%) is ₹190, indicating that the mean is significantly higher than the median.

This difference confirms a right-skewed distribution, where higher-priced products pull the average upward. Additionally, 75% of the products are priced below ₹359, whereas the maximum sale price reaches ₹12,500.

This large gap between the typical price range and the maximum value explains the long right tail observed in the histogram.

These figures indicate that higher sale prices represent premium products within the dataset rather than abnormal or erroneous entries.

Breakdown of Products by Category

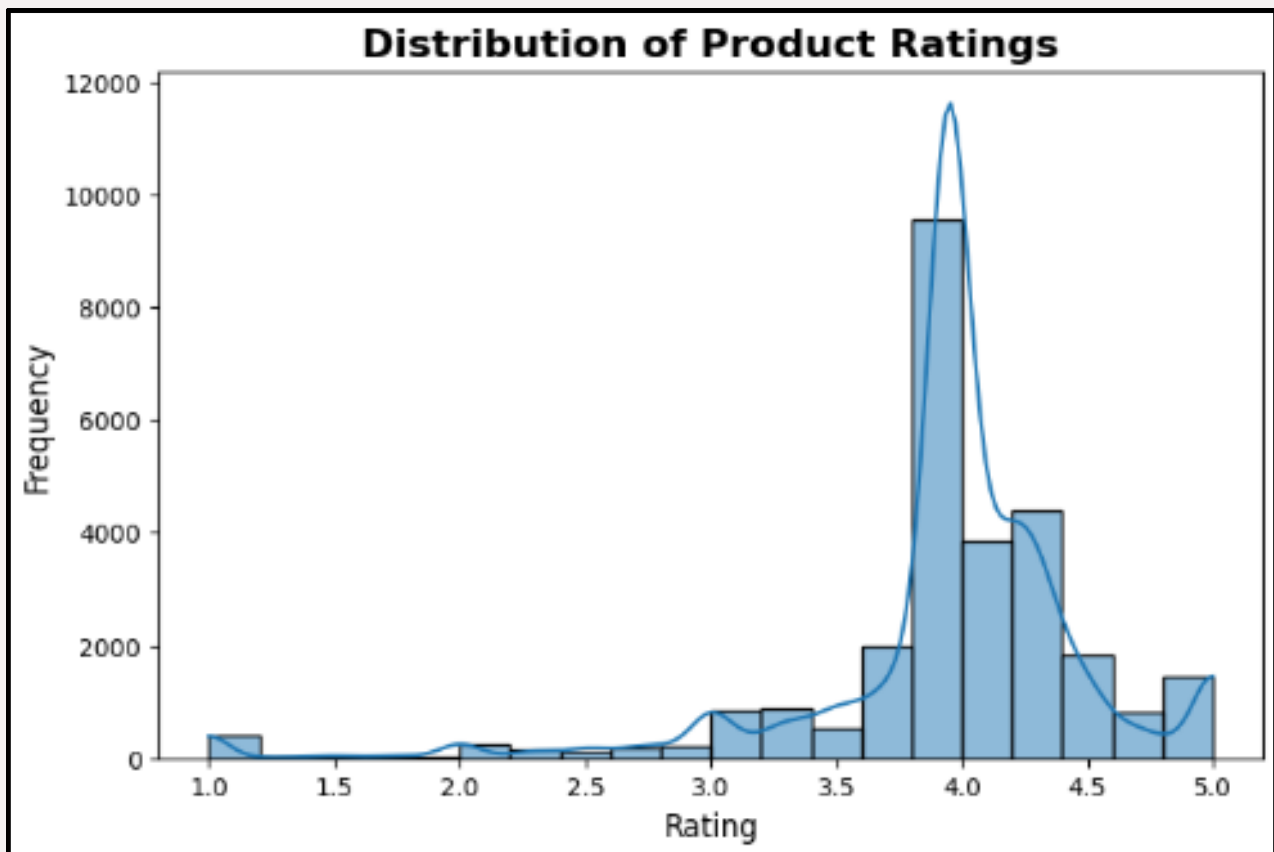


Product distribution is heavily concentrated in Beauty & Hygiene, followed by Gourmet & World Food and Kitchen. This indicates uneven product distribution across segments.

High product concentration in select categories suggests inventory focus and stronger demand presence in these segments, while lower-count categories reflect niche or limited offerings.

This distribution insight supports strategic decisions related to inventory planning, marketing prioritization, and category expansion.

Rating Pattern Across Products



Product ratings are heavily concentrated between 3.5 and 4.5, with a clear peak around 4.0, indicating a strong central clustering of positive feedback.

The distribution shows a mild left skew, as low ratings are relatively rare. This confirms that poorly rated products are minimal, while the majority maintain consistently high customer satisfaction

Such rating stability reflects strong product quality control and positive customer perception across the platform.

Understanding Sale Price vs MRP

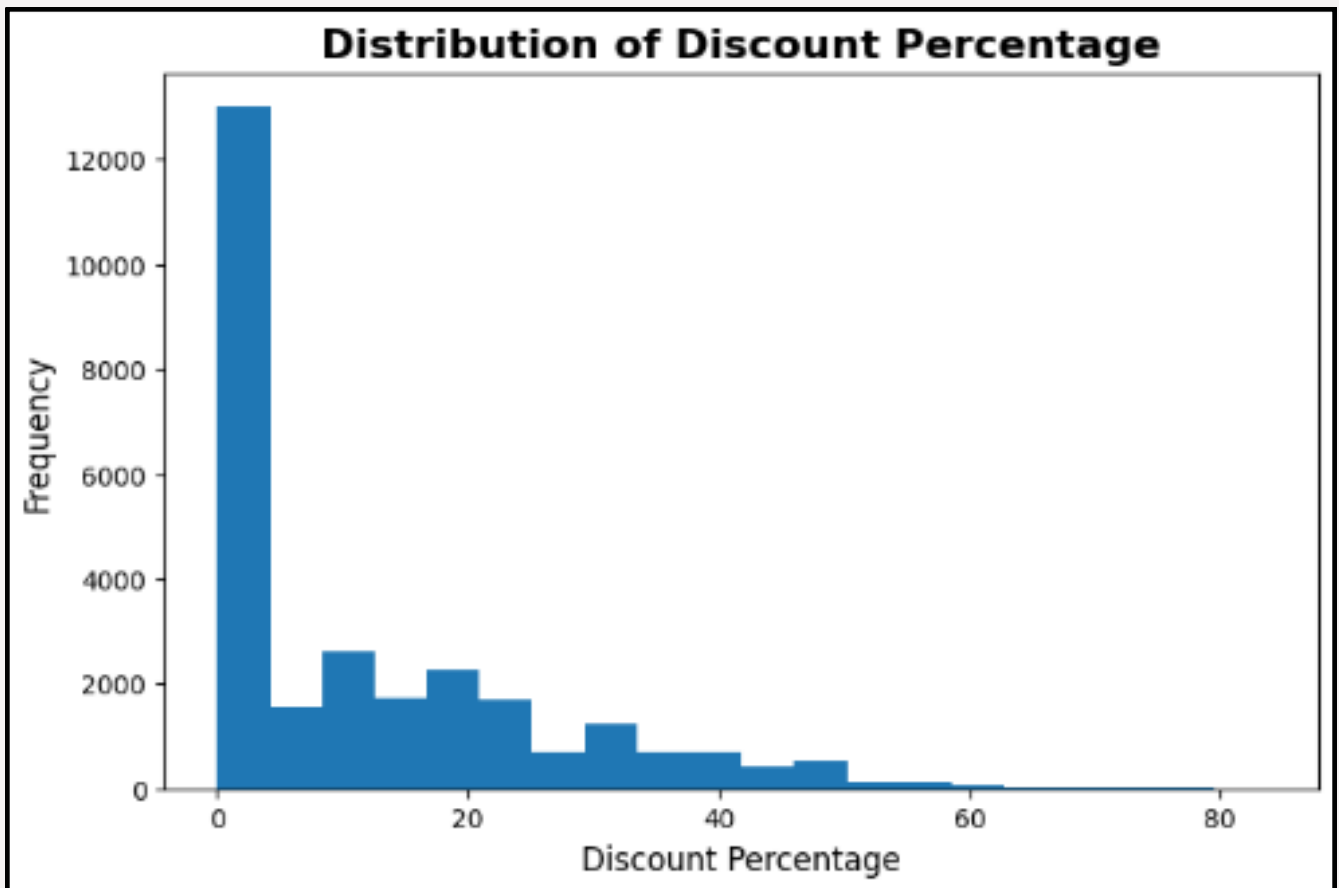


The scatter plot shows a strong positive relationship between Market Price (MRP) and Sale Price, indicating consistent pricing alignment across products.

Most data points lie below the diagonal line, confirming that sale prices are generally lower than MRP and discounts are systematically applied.

The clustered pattern suggests proportional discounting across price ranges, with no signs of irregular or random pricing behavior.

Discount Percentage Analysis

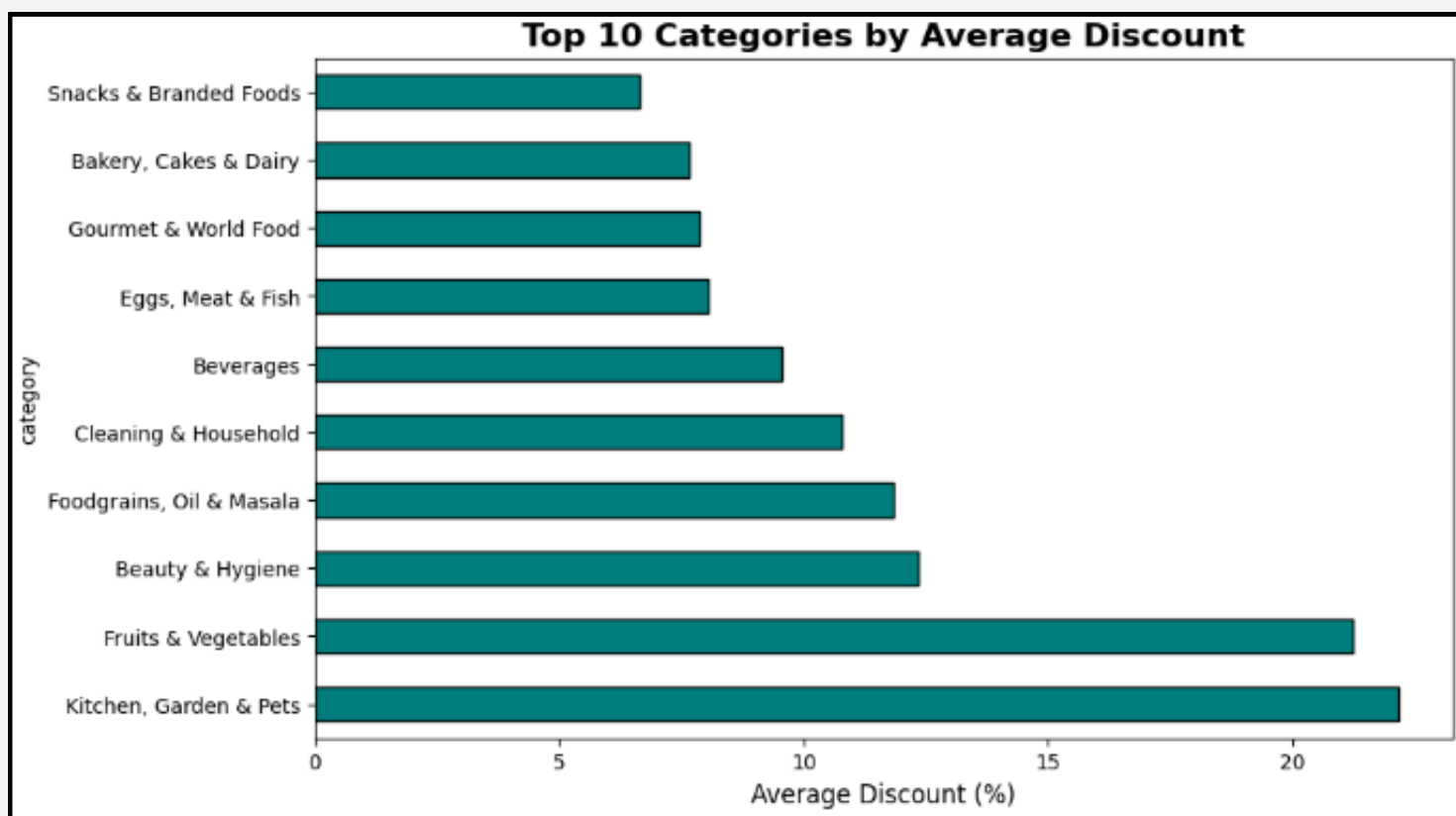


The distribution of discount percentages is heavily concentrated in the lower range, indicating that most products receive minimal to moderate discounts.

As the discount percentage increases, product frequency declines sharply, showing that high-discount items are comparatively limited.

This pattern reflects a controlled pricing strategy where discounts are applied selectively rather than aggressively across the entire catalog.

Average Discount Across Product Categories



The bar chart compares average discount percentages across the top 10 product categories. Kitchen, Garden & Pets and Fruits & Vegetables show the highest average discounts, indicating more aggressive promotional pricing in these segments.

In contrast, categories such as Snacks & Branded Foods and Bakery, Cakes & Dairy maintain comparatively lower average discounts, suggesting more stable pricing structures.

The variation confirms that discount intensity differs significantly by category rather than being uniformly applied across the platform.

Conclusion

This exploratory analysis reveals a structured and strategically aligned pricing ecosystem across the platform. Market Price and Sale Price demonstrate strong consistency, confirming systematic discount application rather than arbitrary pricing behavior.

Discount patterns vary meaningfully across categories, indicating targeted promotional strategies rather than uniform markdown policies. Higher discount intensity in select segments reflects competitive or demand-driven positioning.

Outlier examination confirms that extreme values represent premium products rather than data anomalies, preserving dataset integrity. Additionally, the strong concentration of ratings in the higher range suggests overall positive customer perception.

Overall, the findings highlight controlled pricing mechanisms, category-focused discount strategies, and stable product quality — insights that can support data-driven decisions in pricing optimization, promotional planning, and inventory strategy.

Thank You

Thank you for reviewing this analysis.

I welcome any feedback or suggestions for
further improvement.
